



LENDING CLUB CASE STUDY SUBMISSION

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Lending Club Case Study Analysis

- **Business Information:**

- Lending club is a marketplace for personal loans that matches borrowers who are seeking a loan with investors looking to lend money and make a return.
- Borrowers can easily access to lower interest rate loans through a fast online interface.

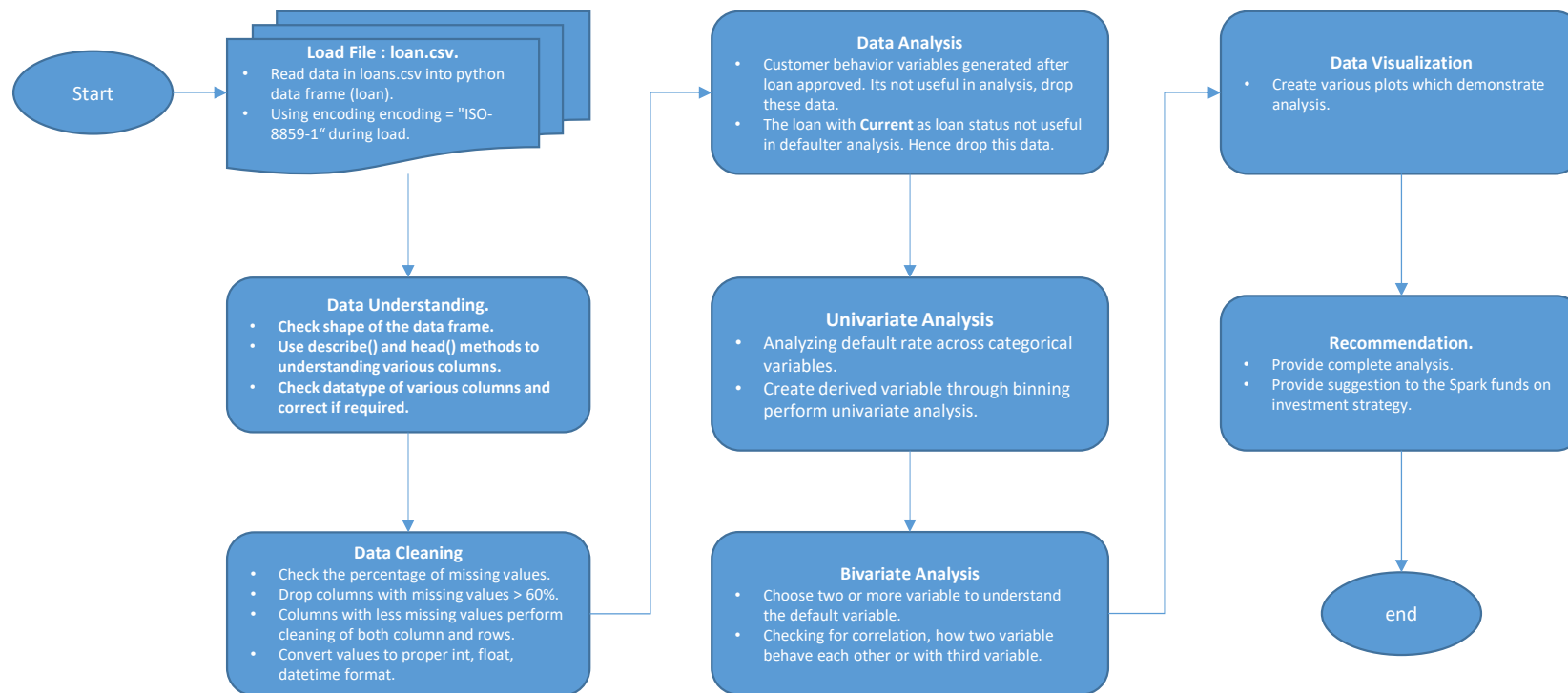
- **Objective:**

- Identify variables which are strong indicators (driver Variables) of loan default.
- Lending Club can use those variable for approval/rejection of loan application.

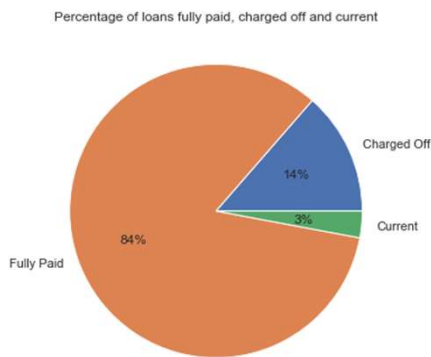
- **Strategy:**

- As a data scientist working for Lending Club analyze the dataset of previous loan applicants using EDA to identify attributes which all attributes influence in default.

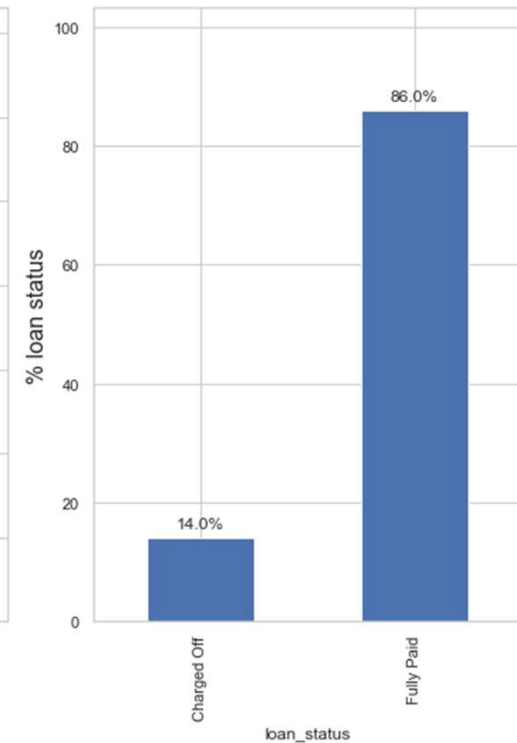
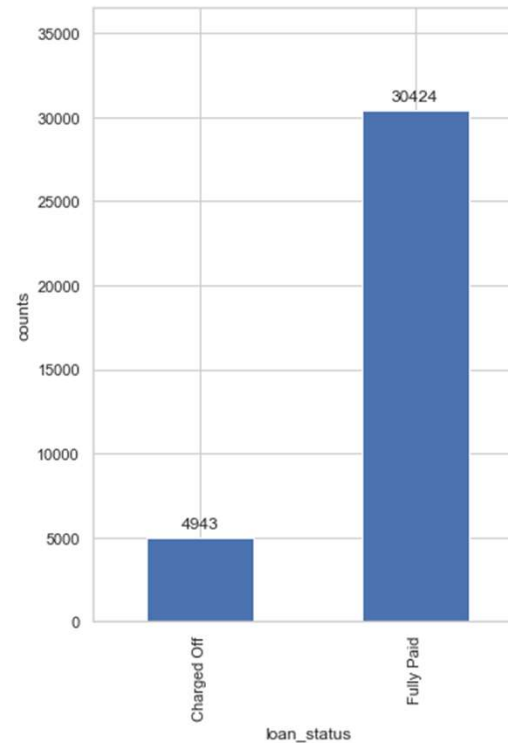
Flow Chart



Loan Status Analysis

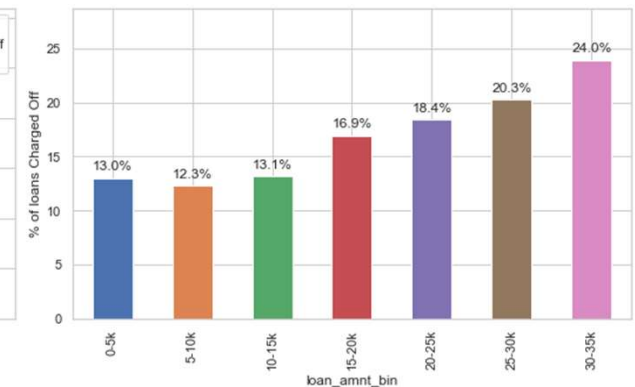
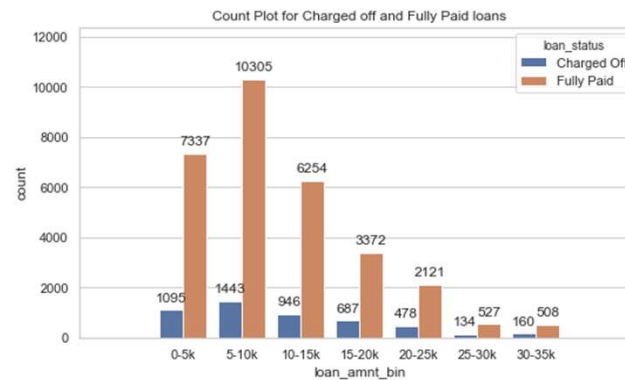
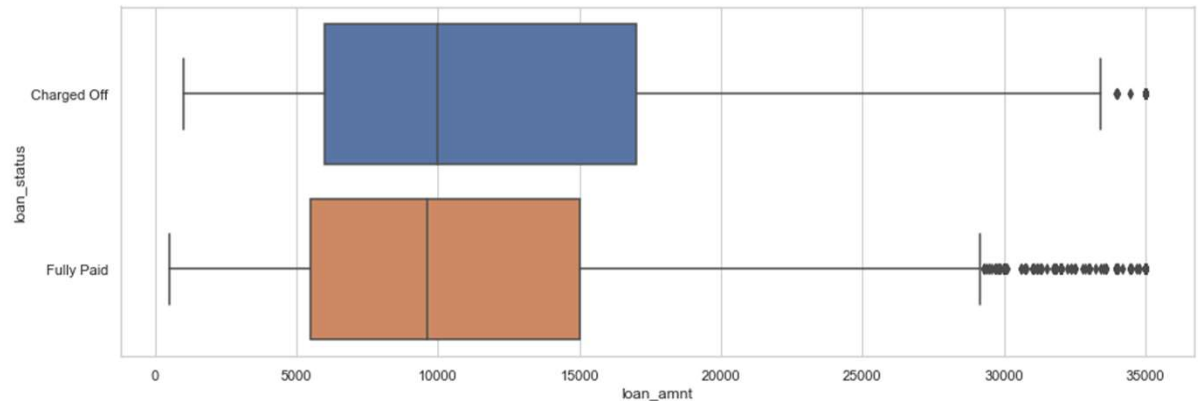


- Fully Paid : 84%, Charged Off : 14% and Current : 3%.
- It is observed that after removal of 'Current' loans, 86% of loan is fully paid, whereas Charged Off is 14%.



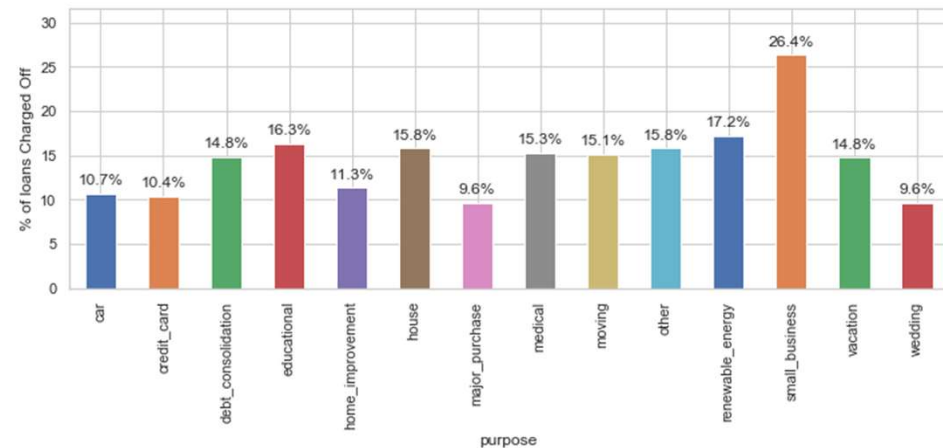
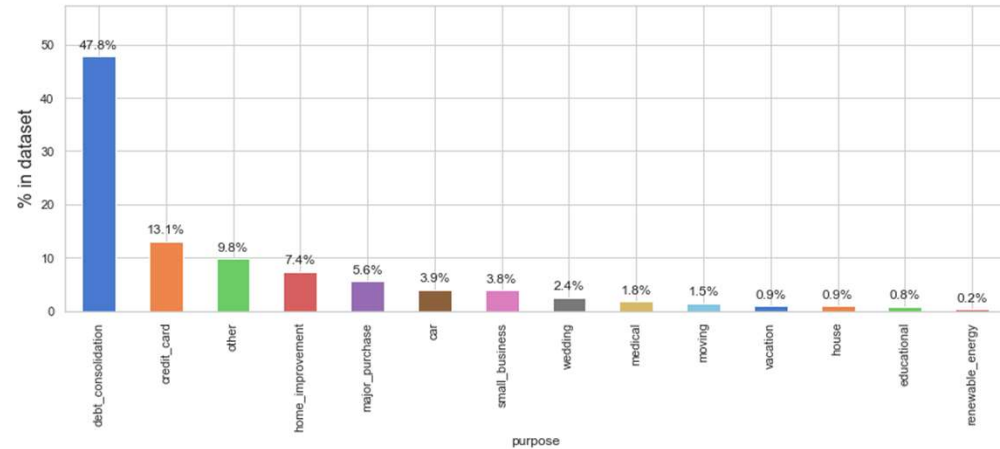
Loan Amount Analysis

- Average loan amount in Charged Off are at higher side.
 - Fully Paid - mean: 10945.73 and median:9600.00
 - Charged Off - mean:**12286.15** and median:10000.00
- Most of the loans were below 20K.
- Percentage of loans "Charged Off" are continuously increasing as loan amount increases.
- The loans in the range 25-30K and 30-35K are lesser in number but carries high risk of becoming default.

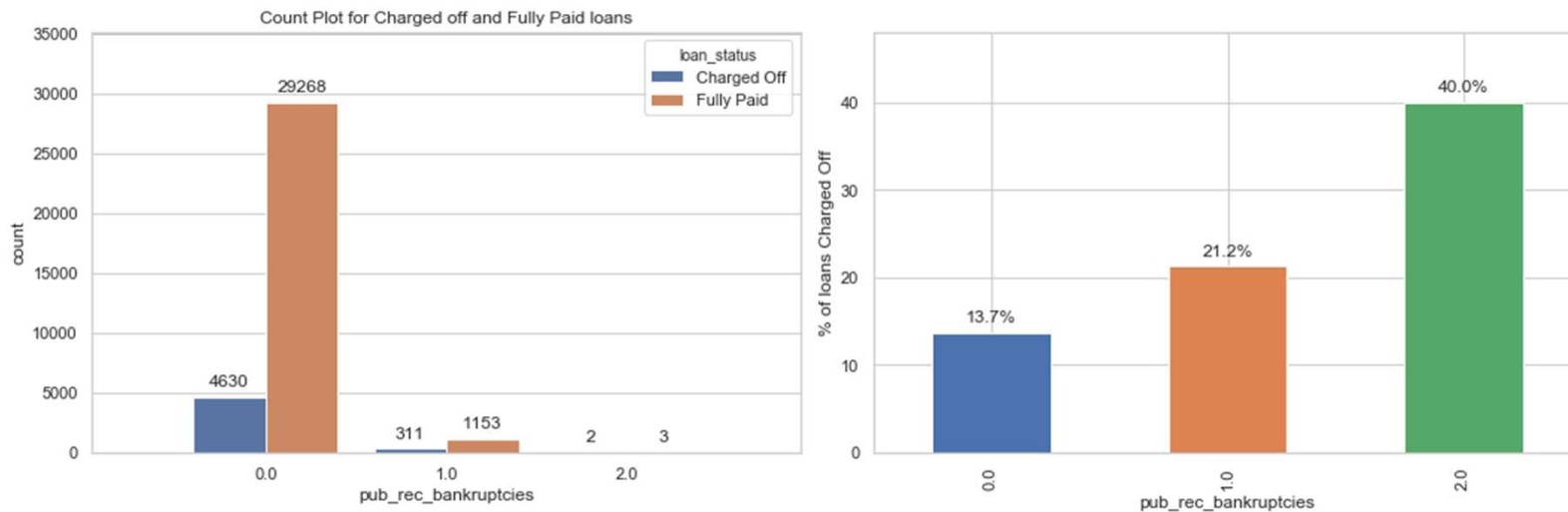


Loan Purpose Analysis

- About 47.8% of the loans are issued for the purpose of dept_consolidation.
- 26.4% of the loans for small_business are charged Off, need to consider this while lending loan.
- 17% of the loans for renewable_energy are charged Off, but the number is too less to be of significance.
- More than a quarter of loans taken for the purpose of running small business see defaults.



Bankruptcy Record Analysis

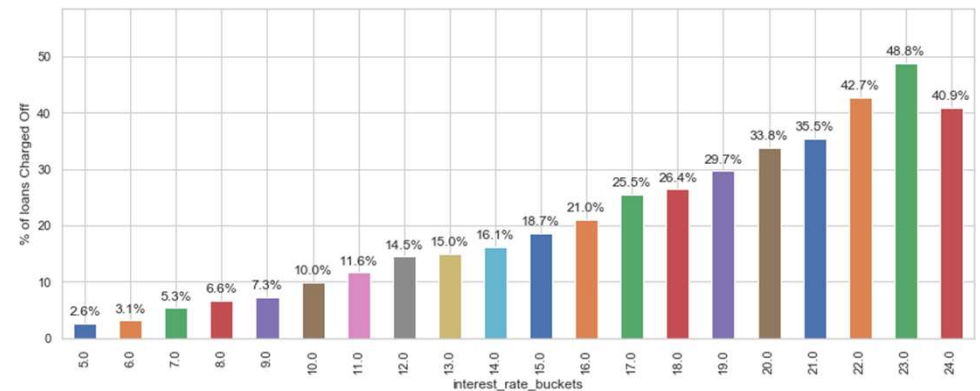
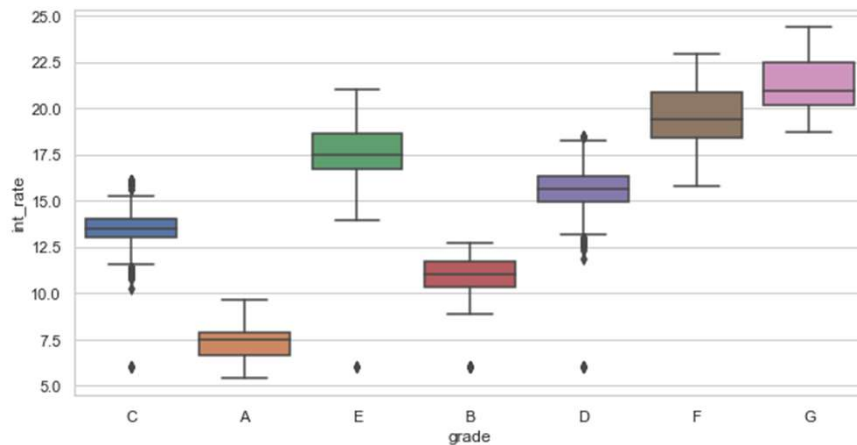
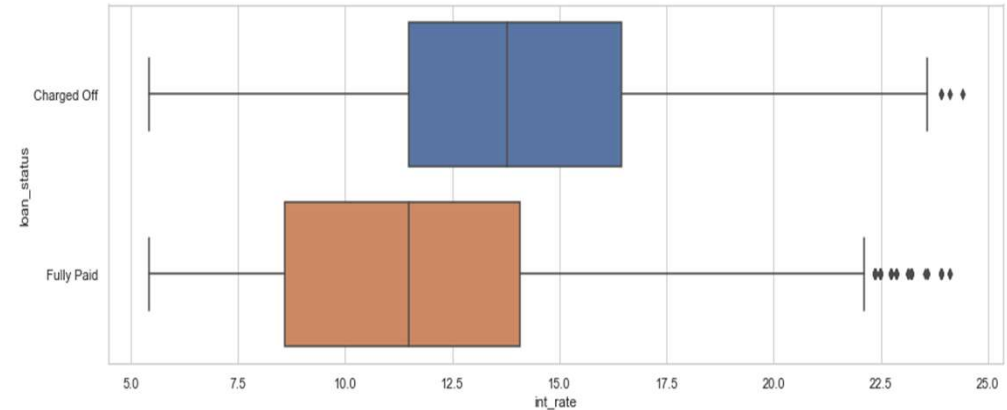


- The % of loans "Charged Off" with previous history of bankruptcies.
 - Previous history of One-time bankruptcy is 21.2%.
 - Previous history of Two-time bankruptcy is 40.0%.

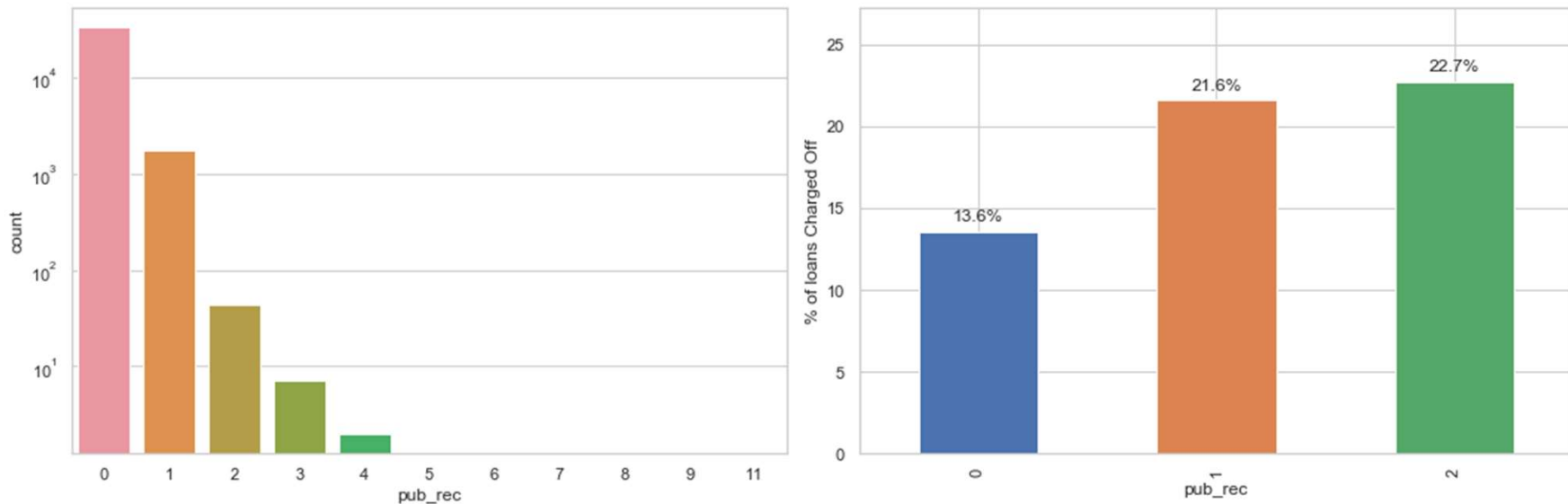


Interest Rate vs Grade Analysis

- The interest rates varies from 5.42% to 24.4% with average interest rate of 11.8%.
- The interest rate for Charged Off loans are higher than for Fully paid.
- As risk increases the rate of interest imposed on the loan also increases.
- Lending Club is correctly charging higher interest rates for lower grade loans.



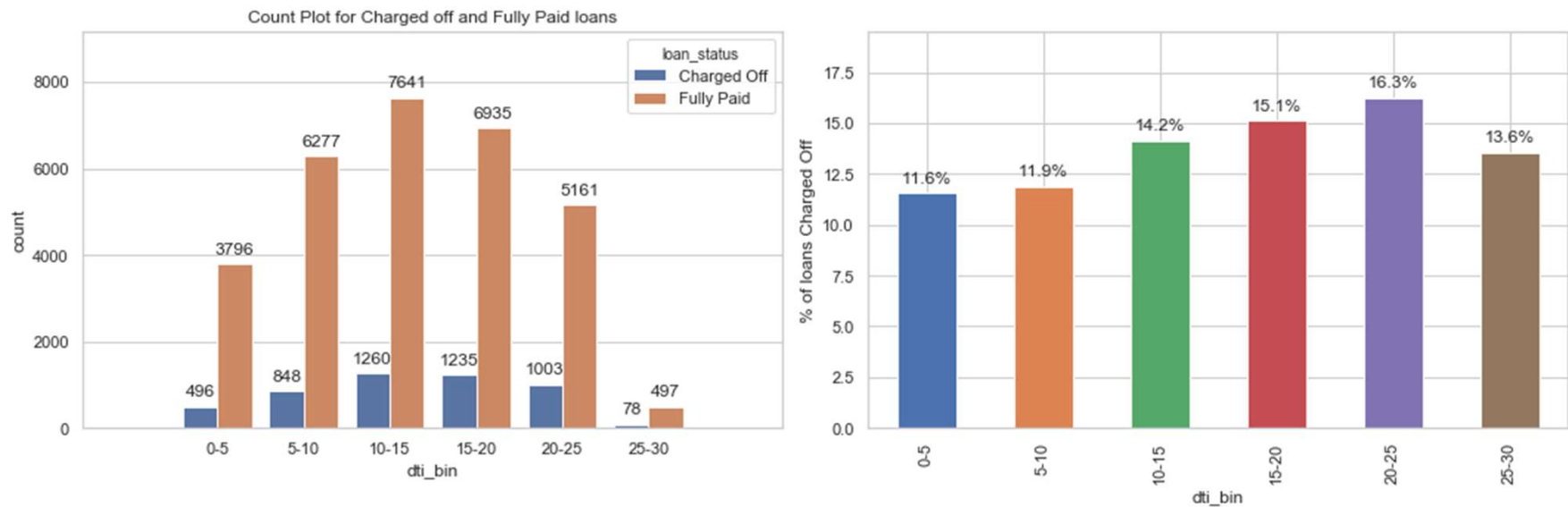
Analysis of Public Derogatory Records



- Around 95% of the borrowers have no Public derogatory records and 5% have 1 derogatory record.
- Even having 1 derogatory record substantially increases the chances of getting default.

Note: A derogatory item is an entry that may be considered negative by lenders because it indicates risk and hurts your ability to qualify for credit or other services. Public records and collections are derogatory items because they reflect financial obligations that were not paid as agreed.

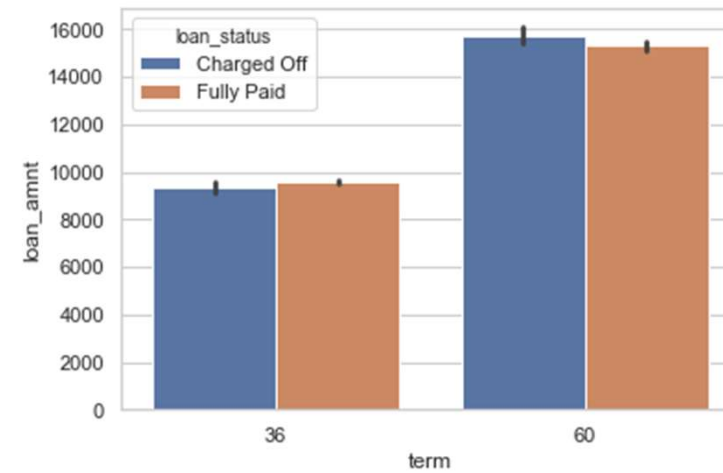
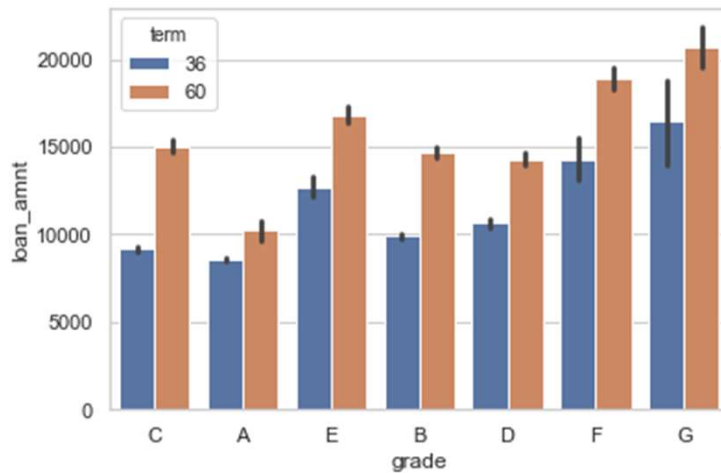
Analysing based on dti



- Higher the dti the chances of loan being defaulted increases.

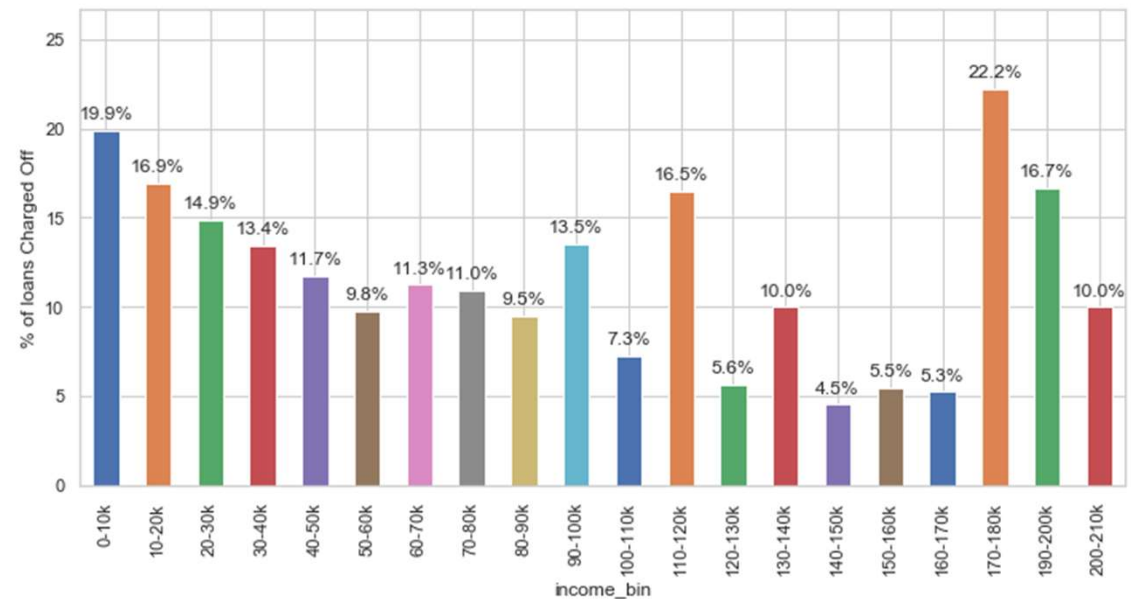
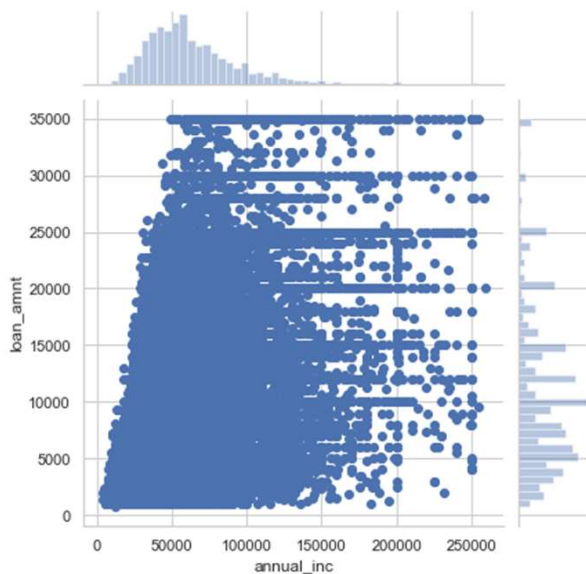
Note : $dti = \frac{\text{Borrower's total debt}}{\text{Borrower's total income}}$

Term vs Grade vs Loan Amount



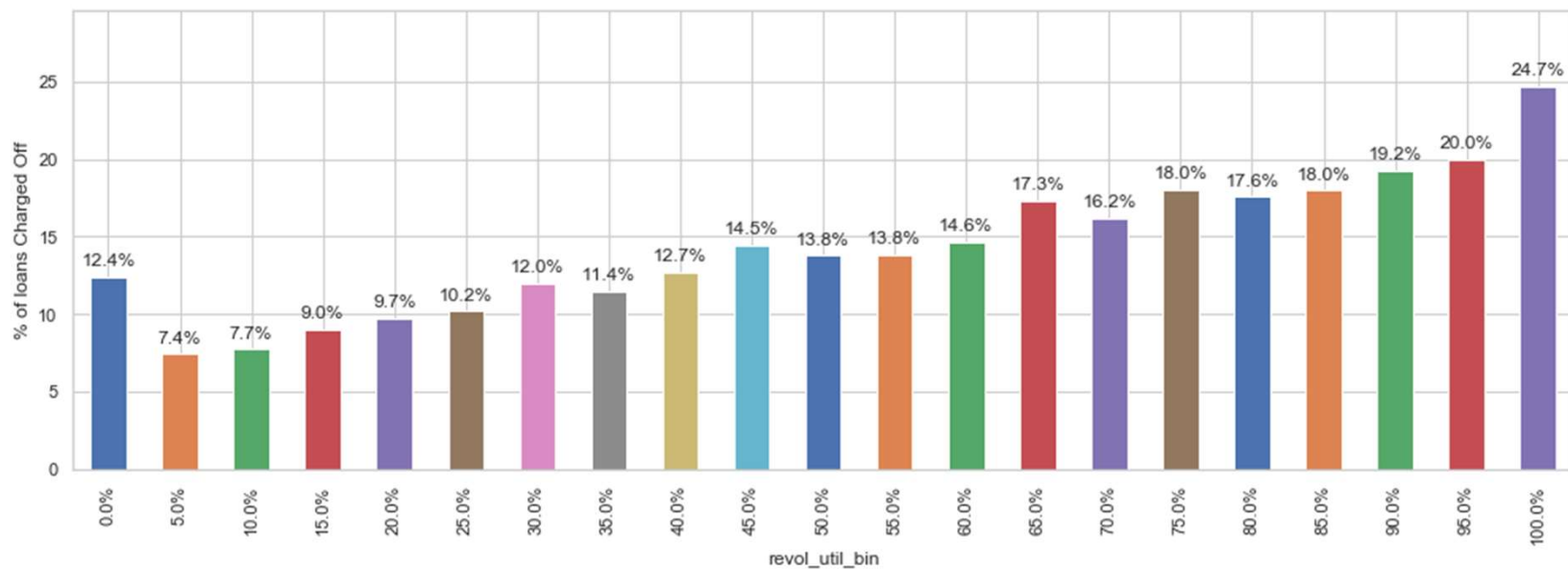
- Higher loan amounts are associated with longer terms and increases defaulter rate.
- Higher loan amounts are associated with lower grade and longer tenure.

Analysing Ratio of amount of Income with Charged Off



- People who are early stage of their career with income in the range 0-10K charged off more, about 19.9%.
- People with income range 170-180K bin have got maximum Charged Off, about 22.2%.
- Borrowers with average income lower than 50000 are taking loans of 25000 or higher.

Analysis of Revolving Line Util Rate



- Applicants with High revolving line credit utilization rate results in more defaulters.



Recommendations

Minimise the loan approval for small businesses and increase the scrutiny to reduce the default rate.

Stop approving loans for applicants with prior history of bankruptcy or bad records.

LC should avoid lending higher loan amount to lower income borrowers.

Start charging higher interest rates for loans with dti greater than 20.

Stop approving higher loan amount to lower grade borrowers for longer tenure.

Lending Club should strictly verify the income of applicant.

Reduce loan approval for high revolving line utilization rate.

